

NSSL

Bulk Payment Functionality

ADDENDUM

23 August 2024

[Version 2.0]

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Document distribution list

NSSL		
Name	Position	Date sent

BPC		
Name	Position	Date sent

Document approval

NSSL			
Name	Position	Signature	Date

BPC Banking Technologies			
Name	Position	Signature	Date
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Revision history

Version	Date	Summary of changes	Affected sections	Author
0.1	23.08.2024	Initial Draft	All	Avinash Arrah

Glossary

Term	Definition
PP	Participant Portal
XML	Extensible Markup Language
CSV	Comma separated Values
API	Application Programming Interface
ACSC	AcceptedSettlementComepletedDebitorAccount
ACSP	AcceptedSettlementInProgress
ACSC	AcceptedSettlementComepletedDebitorAccount
SVIPS	SmartVista Instant Payment Solution

1. Introduction

1.1. Background and business objectives

A bulk transaction allows you to transfer funds to multiple credit accounts under the same beneficiary participant

Bulk payments are incredibly useful for financial processes like:

- Salary payment
- Pay Dividends
- Humanitarian aids
- Pay Contractors and freelancers

1.2. Scope

- Account verification through signed XML file
- Bulk payment processing initiated through Signed XML file
- Convert and debulk CSV file to XML file

1.3. Assumptions and constraints

- Above functionalities will not impact existing functionalities.
- No new reports will be developed/amended by BPC within the scope of the current document

1.4. Limitation and constraints

- One batch file will contain instructions for only one beneficiary participant.
- One batch file will contain a maximum of 10,000 instructions
- A maximum of 10(configurable number based on overall performance of the available infrastructure) batches can be uploaded on the system, however at a given moment, only one batch can be in processing mode. Hence batch will be processed sequentially.

2. Bulk Payment Processing

2.1. Account Verification Process

2.1.1. High level flow

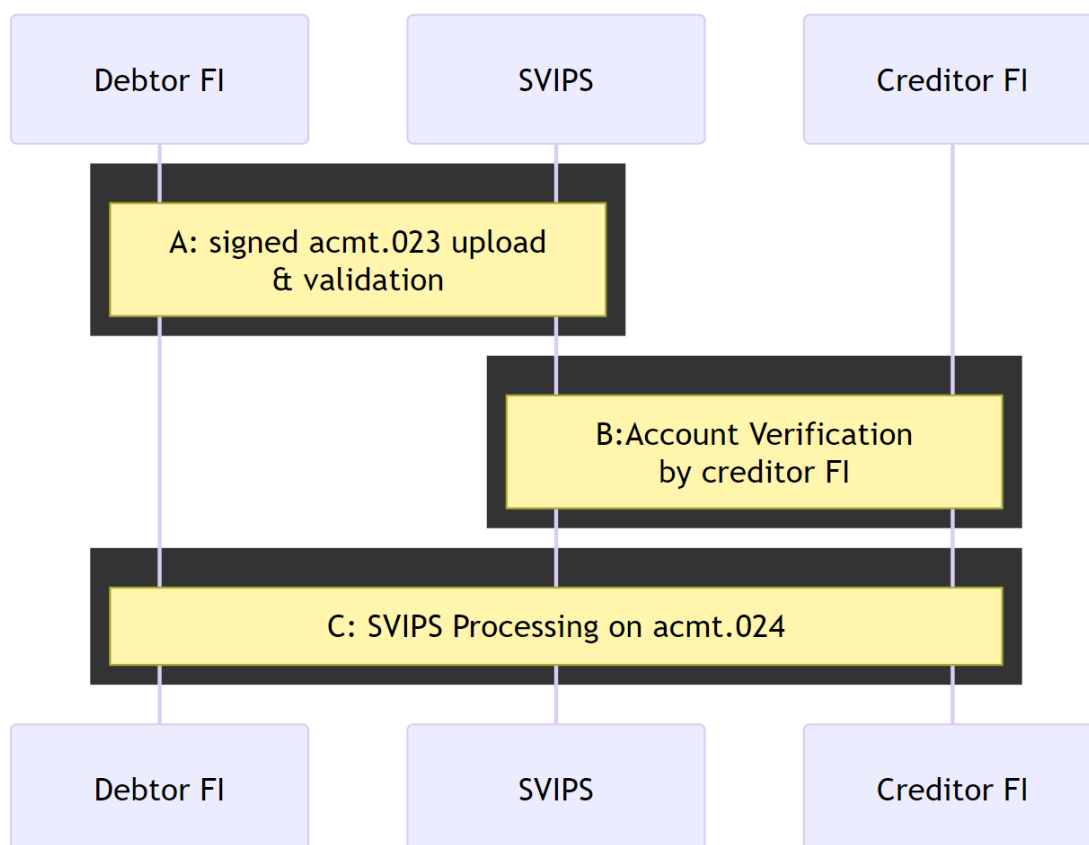


Figure 1 Account Verification process

Sub-Process Reference	Description
A. Signed acmt.023 upload and validation	The Debtor FI uploads a signed acmt.023 (For one beneficiary BIC). The acmt.023 file is submitted to technical and business validation check
B. Account Verification by Creditor Fi	After successful Verification checks, the file is staged on the creditor FI side. Creditor FI validates the files. Creditor FI sends an acmt.024 file containing the status of verification for each account.
C. SVIPS Processing	Upon receiving the acmt.024 file, SVIPS validate the file. In case of validation failure, a reject file is sent to the Creditor FI In the case of validation success, the acmt.024 file is staged on the Debtor FI side.

2.1.2. Process A: Signed acmt.023 file uploaded by Debtor FI & Validation

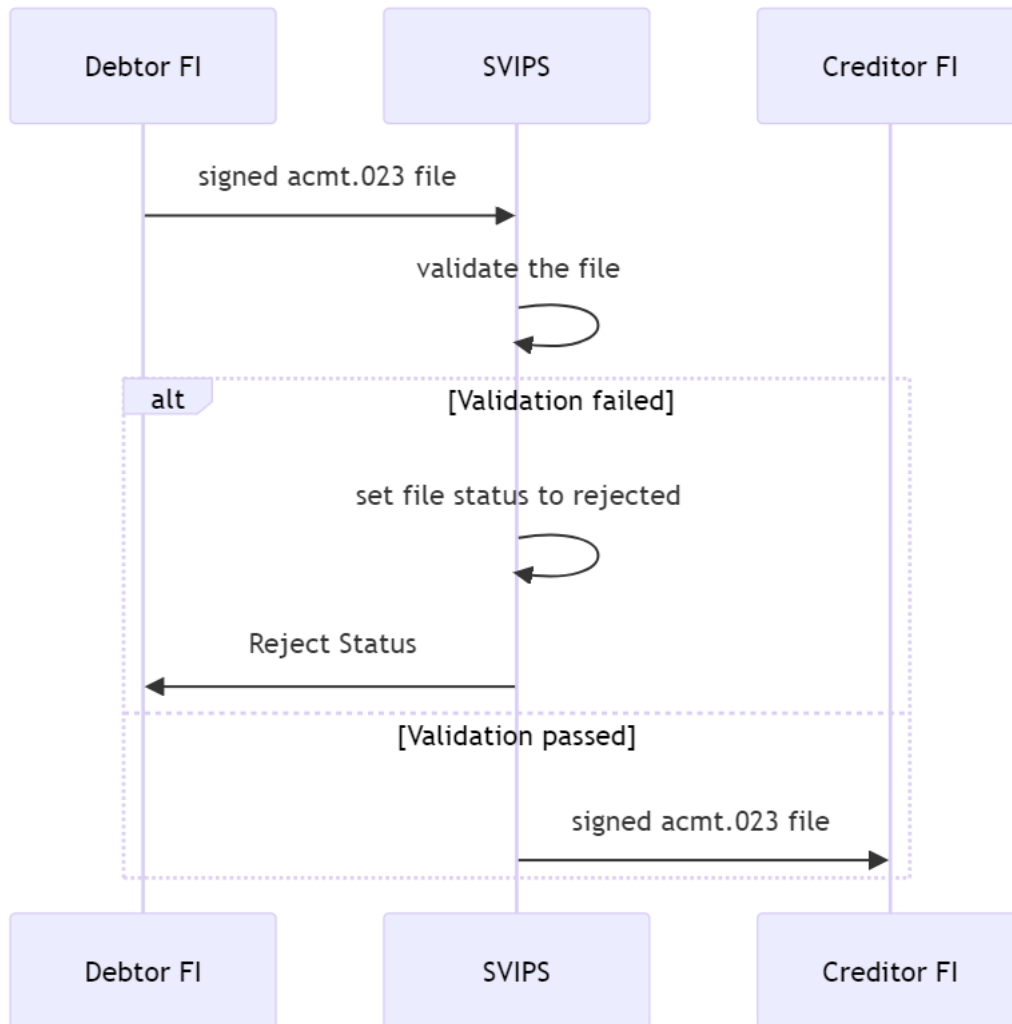


Figure 2 Signed acmt.023 file uploaded by Debtor FI & Validation

2.1.2.1. Validation Checks

Two levels of check will be carried out on the pacs.008 file:

Technical validation

- msg id check for duplication
- xsd validation
- signature verification
- other technical checks similar to all iso20022 xml messages

Business validation

- Fraud checks
- Only one beneficiary BIC per batch
- Currency check.

In the case the file validation fails, the status of the batch file will be set to rejected on Participant portal

2.1.3. Process B: Account Verification process by Creditor FI

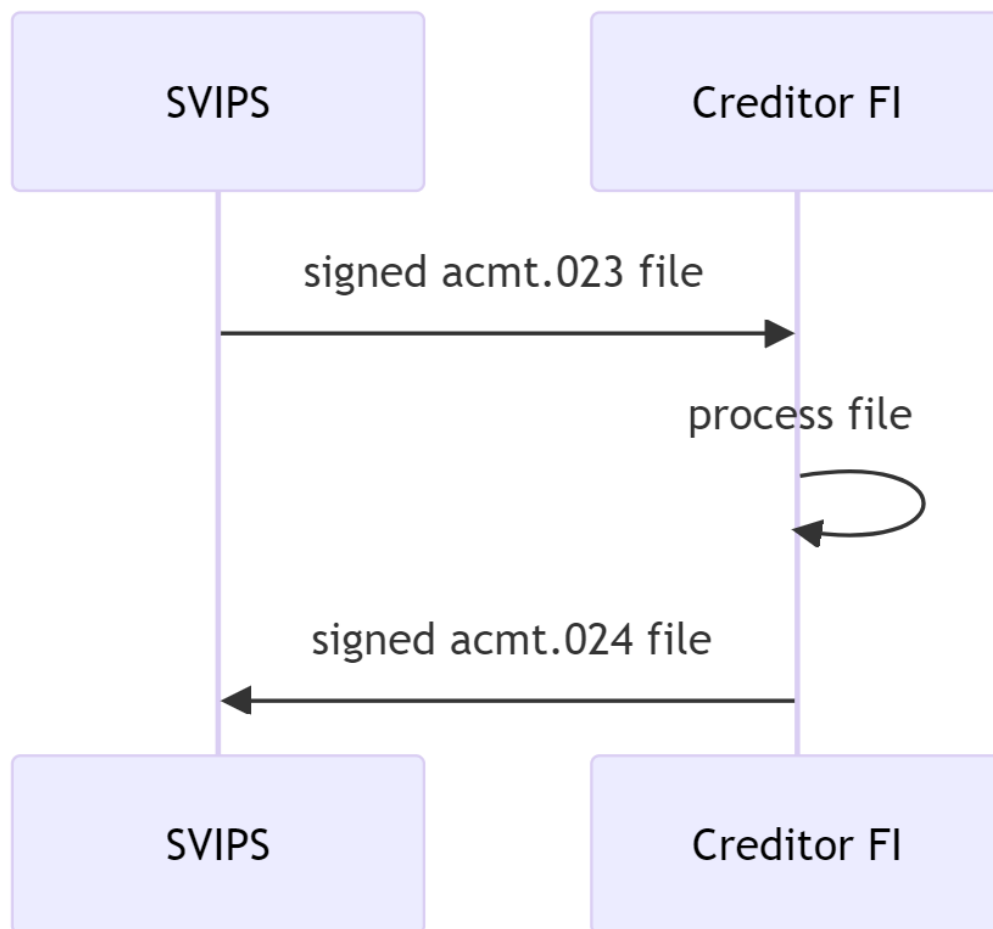


Figure 3 Process B: Account Verification process by Creditor FI

Each instruction on the acmt.023 file will be verified for the following

- Validate Account Existence
- Check Account status
- Check Allowed transactions
- Check Account currency

2.1.4. Process C: SVIPS processing on acmt.024

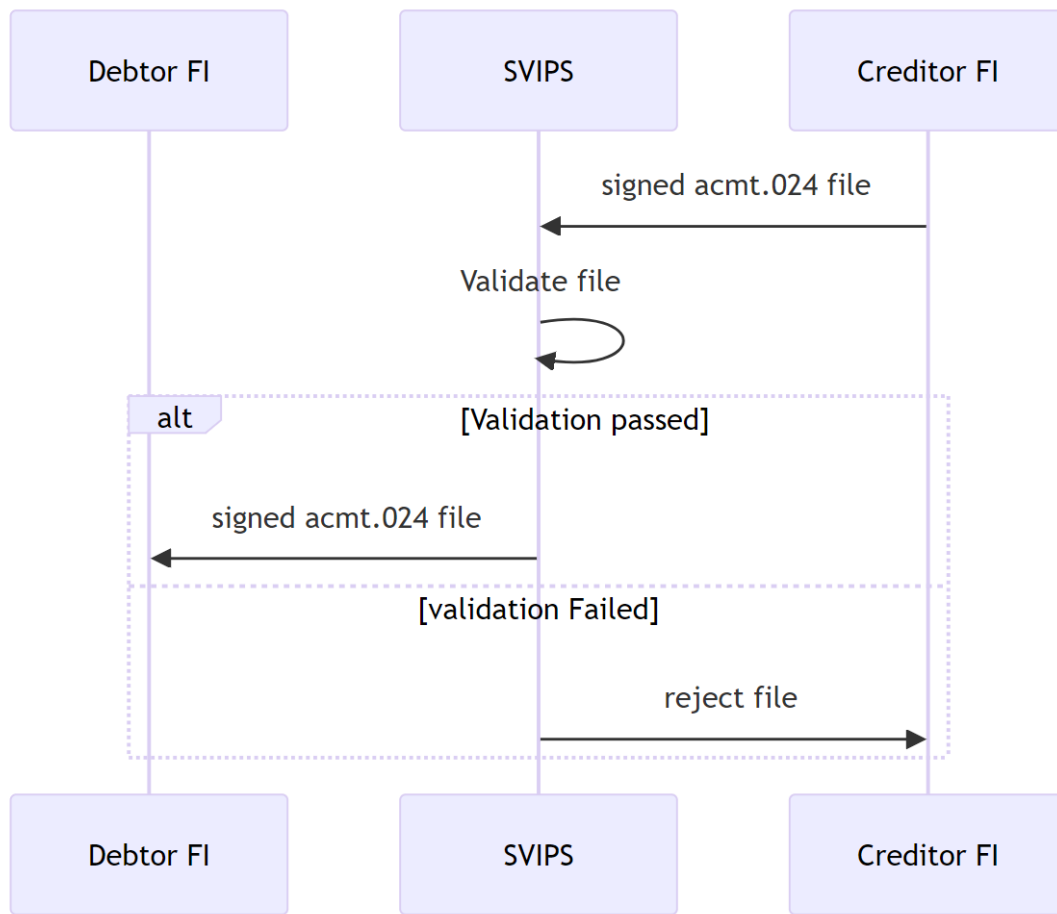


Figure 4 Process C: SVIPS processing on acmt.024

2.1.4.1. Validation Checks

Two levels of check will be carried out on the acmt.024 file:

Technical validation

- msg id check for duplication
- xsd validation
- signature verification
- other technical checks similar to all iso20022 xml messages

Business validation

- Original Assignment exists

2.2. Bulk Payment Processing through Signed Pacs.008 File

2.2.1. High level Flow

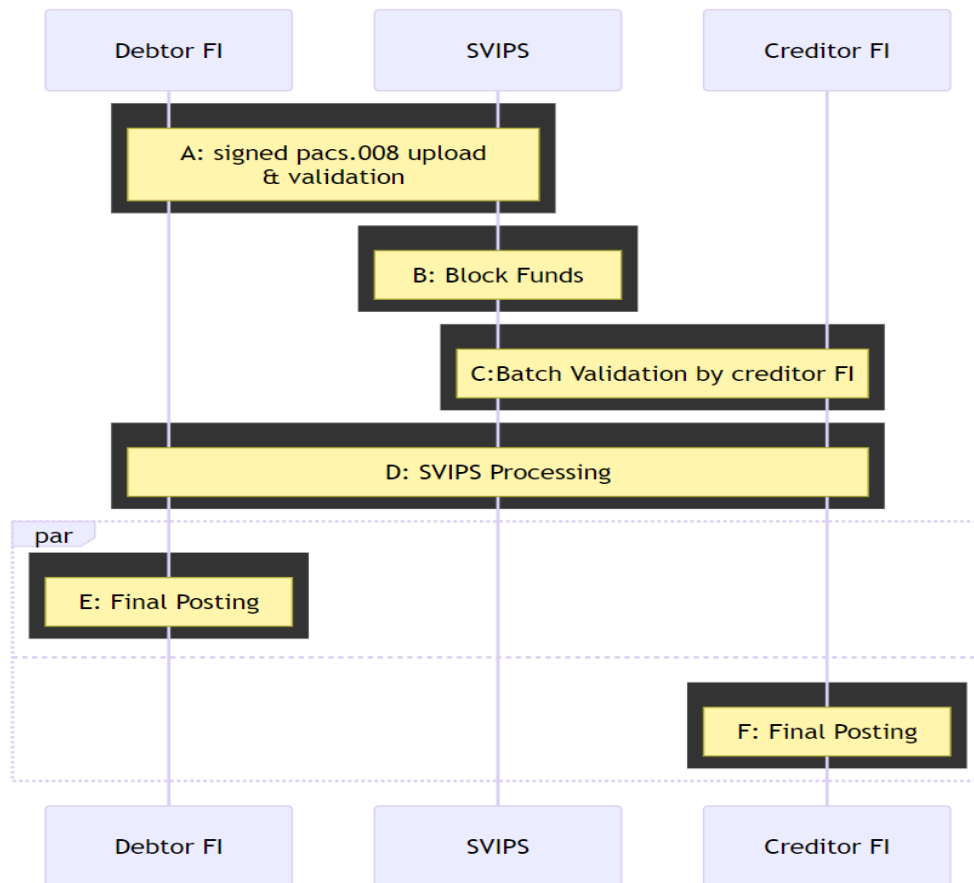


Figure 5 High level Flow diagram

Sub-Process Reference	Description
D. Signed pacs.008 upload and validation	The Debtor FI uploads a signed pac.008 (For one beneficiary BIC). The pacs.008 file is submitted to technical and business validation check
E. Block Funds	After the validation checks, the total amount of the batch is check against available liquidity. If sufficient liquidity, the total amount of the batch is block in the technical counter else, the batch is set to pending not enough liquidity state.
F. Batch validation by Creditor FI	After funds is successfully block, the pacs.008 file is staged on the creditor FI side. Creditor FI validate the file and sends a pacs.002 file with global status RJCT (whole file rejected) or pacs.002 with global status ACSP (Accepted settlement in progress)

Sub-Process Reference	Description
G. SVIPS Processing	In the case of pacs.002 with status ACSP, SVIPS unblock the funds from the technical counter and update net position. Sends pacs.002 with status ACSC to both creditor and debtor FI In the case of pacs.002 with status RJCT, SVIPS unblock the funds from the technical counter and marks the batch as rejected by creditor FI
H. Final Posting on Debtor FI	Case pacs.002 with status ACSC received, posting of accounts is carried out on debtor FI customer
I. Final Posting on Creditor FI	Case pacs.002 with status ACSC received, posting of accounts is carried out on Creditor FI end customer

2.2.2. Bulk Payment Batch State

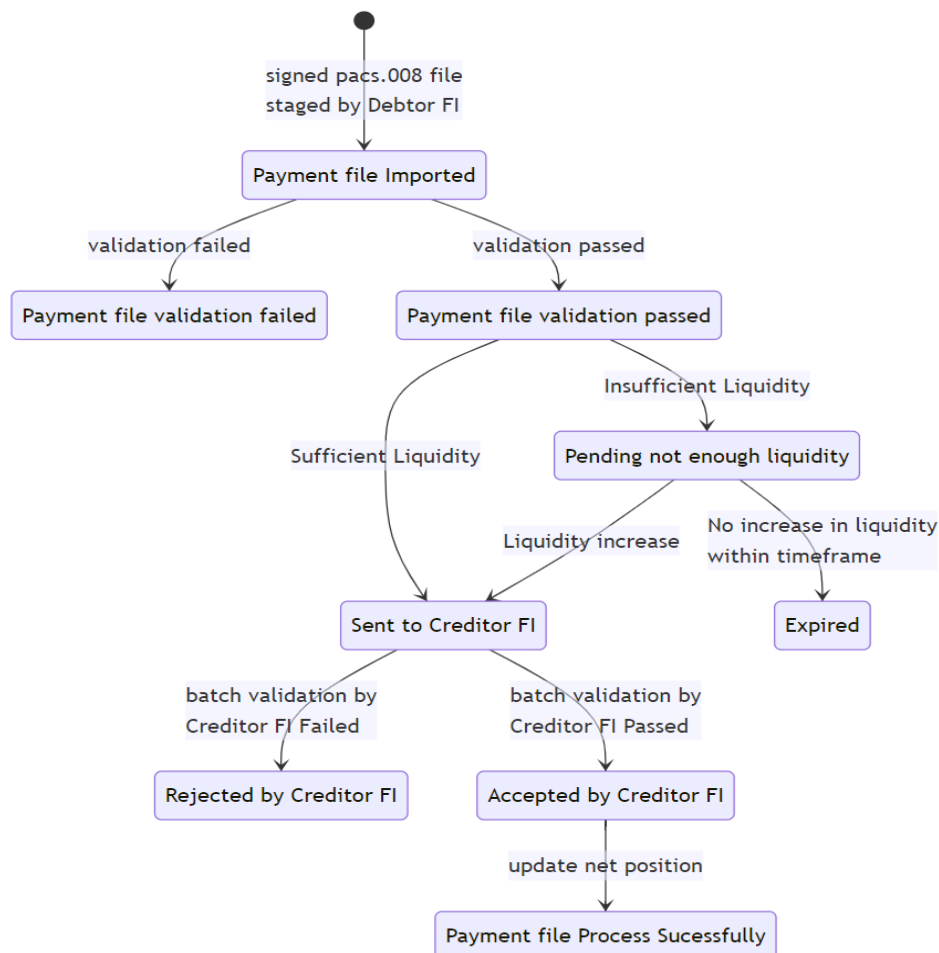


Figure 6 Bulk Payment Batch State

2.2.3. Process A: Upload signed pacs.008 and Validation

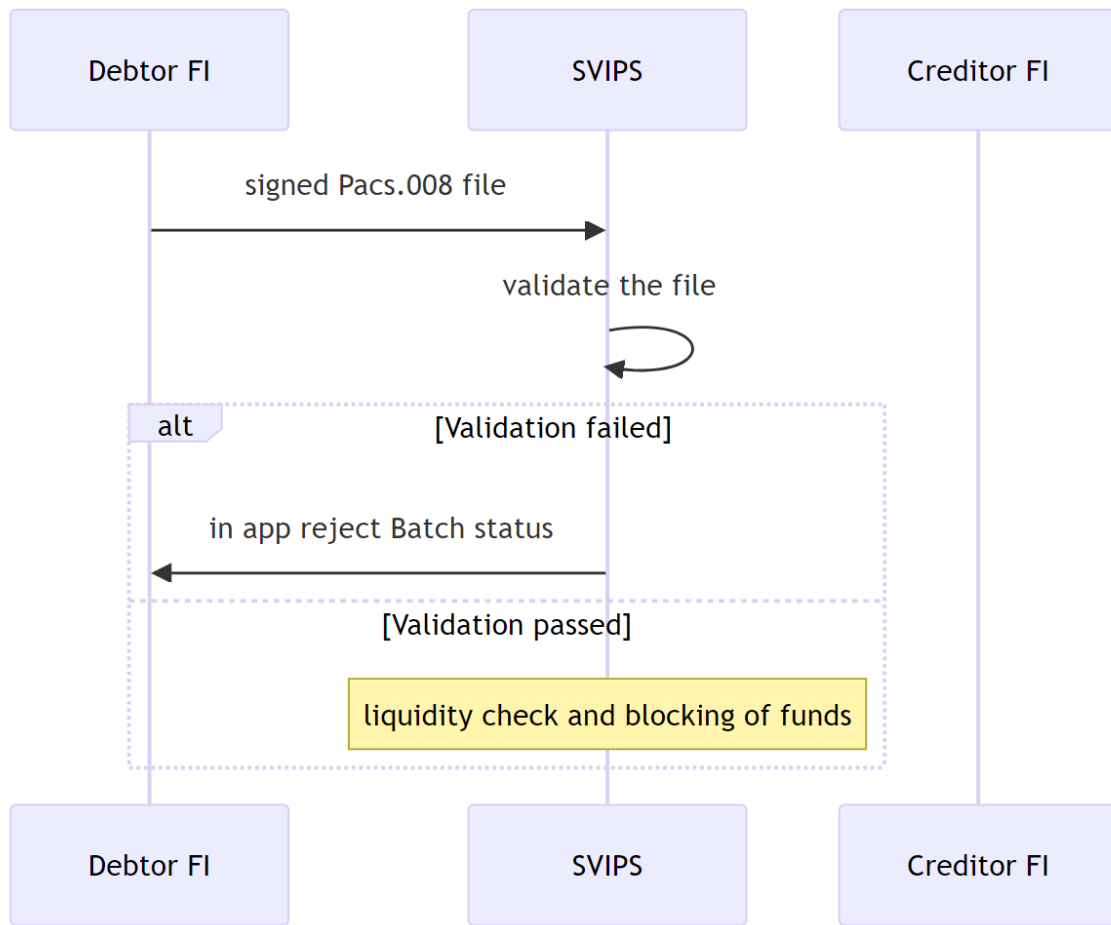


Figure 7 Process A: Upload signed pacs.008 and Validation

2.2.3.1. Validation Checks

Two levels of check will be carried out on the pacs.008 file:

Technical validation

- msg id check for duplication
- xsd validation
- signature verification
- other technical checks similar to all iso20022 xml messages

Business validation

- Fraud checks
- Only one BIC per batch
- Currency check

2.2.4. Process B: Block Funds

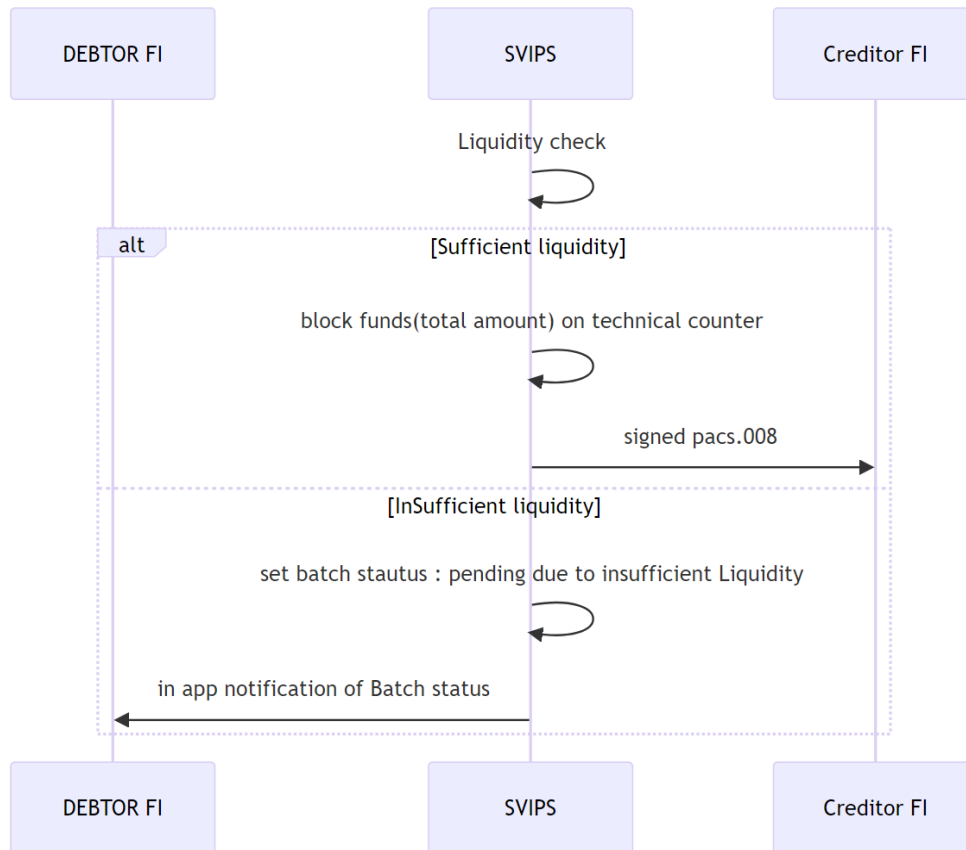


Figure 8 Process B: Block Funds

2.2.5. Process C: Batch Validation by creditor FI

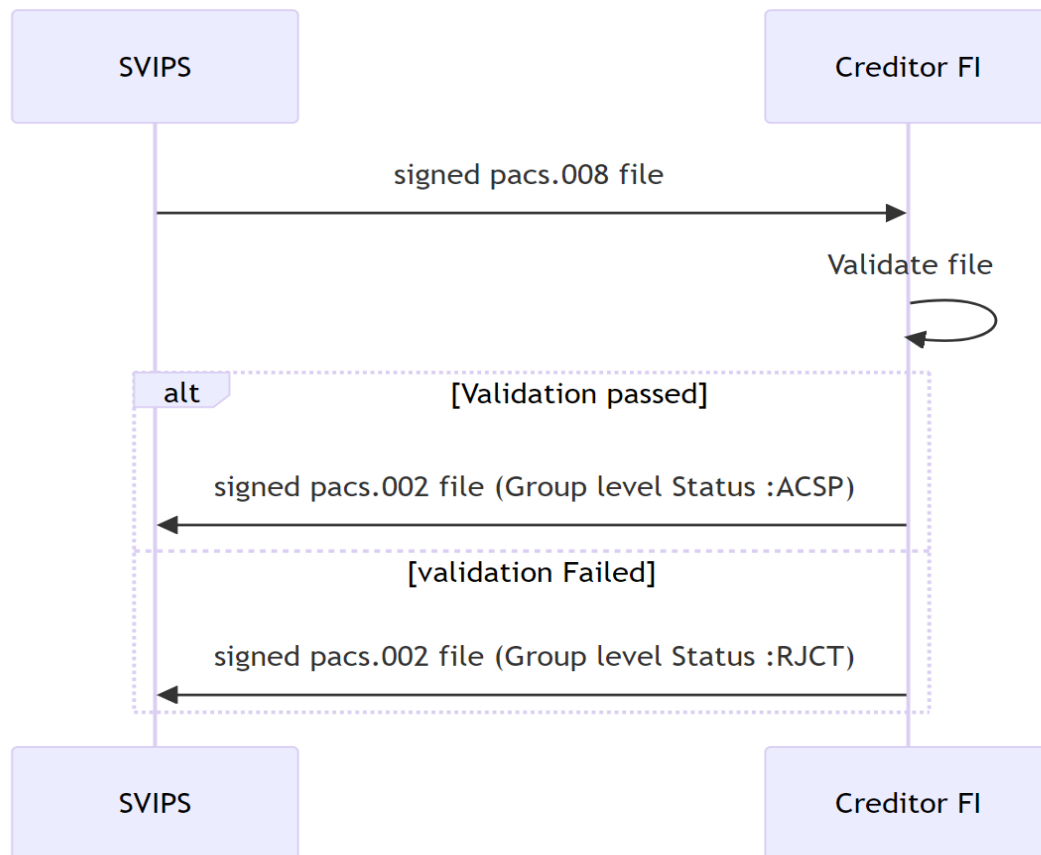


Figure 9 Process C: Batch Validation by creditor FI

Creditor FI will have the possibility to either accepts the whole file or reject the whole file. Thus, the validation status will be sent at Group level.

In the case the validation of all instructions in the file is successful, a pacs.002 with status ACSP (AcceptedSettlementInProgress) is sent to SVIPS.

In the case the validation of any instructions in the file is unsuccessful, a pacs.002 with status RJCT (Rejected) is sent to SVIPS.

Status will be set at Group level

OriginalGroupInformationAndStatus <OrgnlGrpInfAndSts>

GroupStatus <GrpSts>

Individual status of each Instructions will not be sent.

2.2.6. Process D: SVIPS Processing

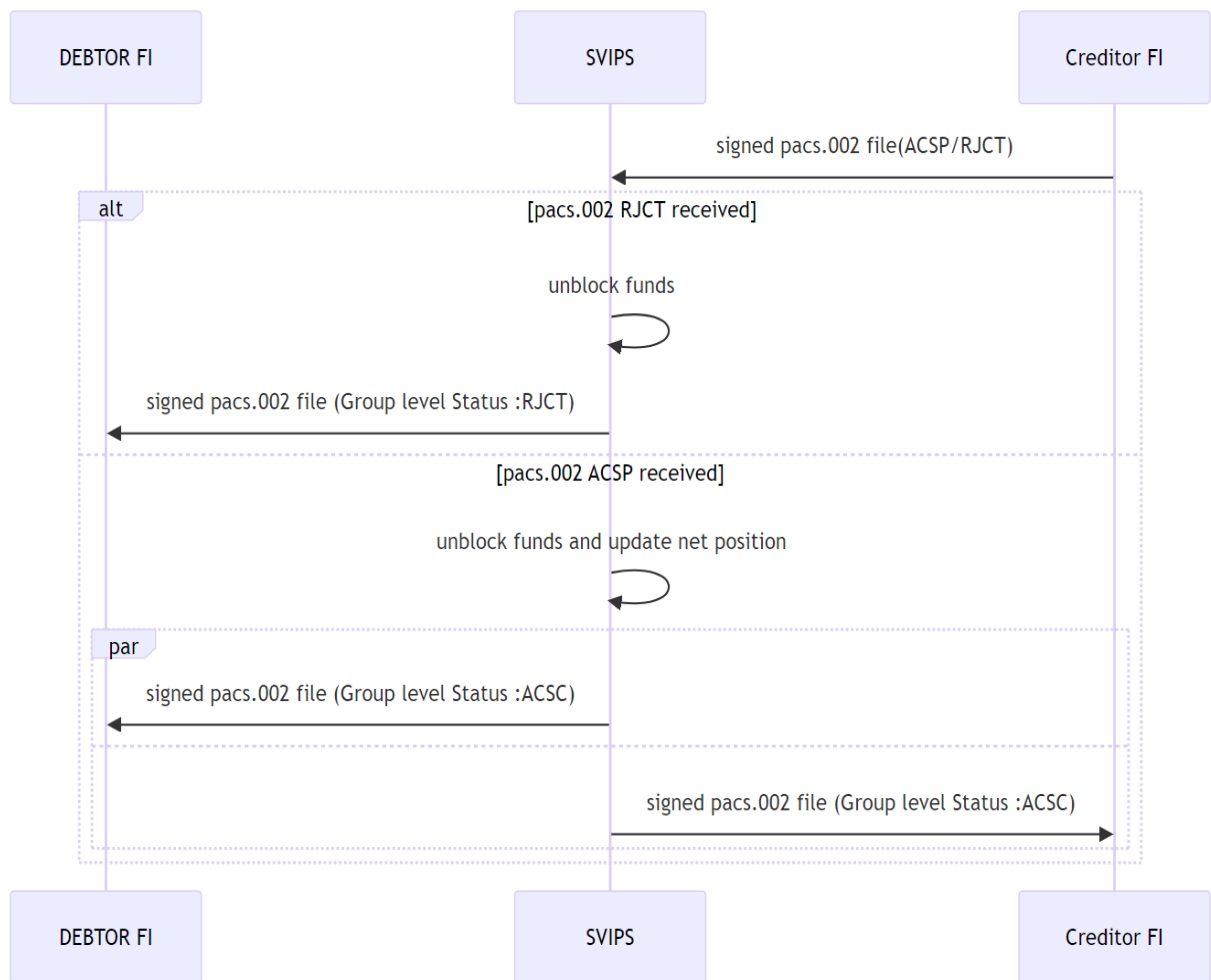


Figure 10 Process D: SVIPS Processing

2.2.7. Process E: Final Posting on Debtor FI Side

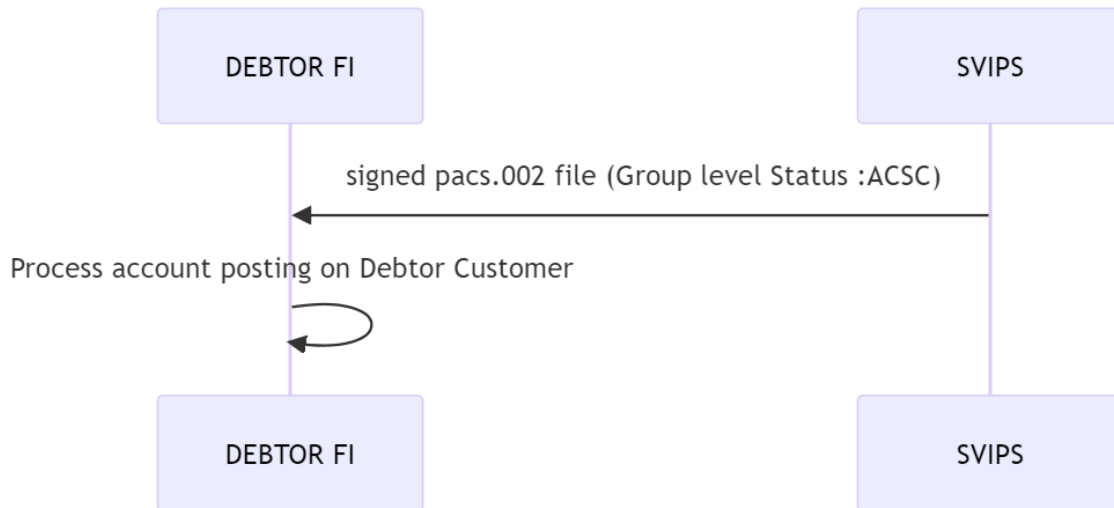


Figure 11 Process E: Final Posting on Debtor FI Side

2.2.8. Process F: Final Posting on Creditor FI Side

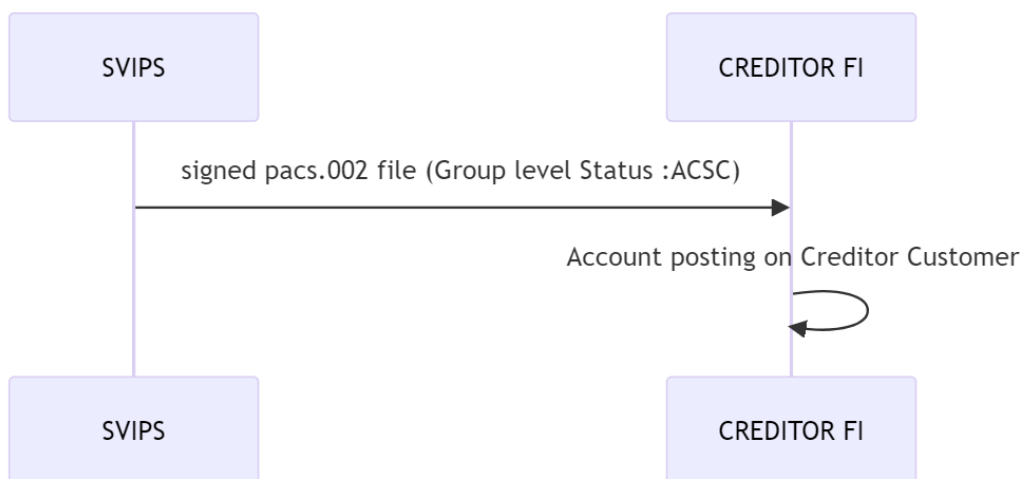


Figure 12 Process F: Final Posting on Creditor FI Side

2.2.9. Pending file Processing due to Insufficient liquidity

Once the original batch file is set in a Insufficient liquidity state, at a configured period (every X mins) and right after SVBO balance file processing we shall check the net debit cap level against the total amount of the pending file. If Sufficient liquidity, then the system may try to process the batch again else the file remains on pending file until expired (configurable bulk payments expiration period) and will get automatically declined.

2.3. Cut off process

The Final RJCT /ACSC pacs.002 will be sent on posting file to SVBO. These records will be part of the reconciliation file to participant. The pacs.002 will contain only group info and no details transactional info.

2.4. Group Status Code Description

ACSC	AcceptedSettlementCompletedDebitorAccount
RJCT	Reject
ACSP	AcceptedSettlementInProgress

Table 1 Status Code Description

2.5. CSV file Processing

2.5.1. Process flow

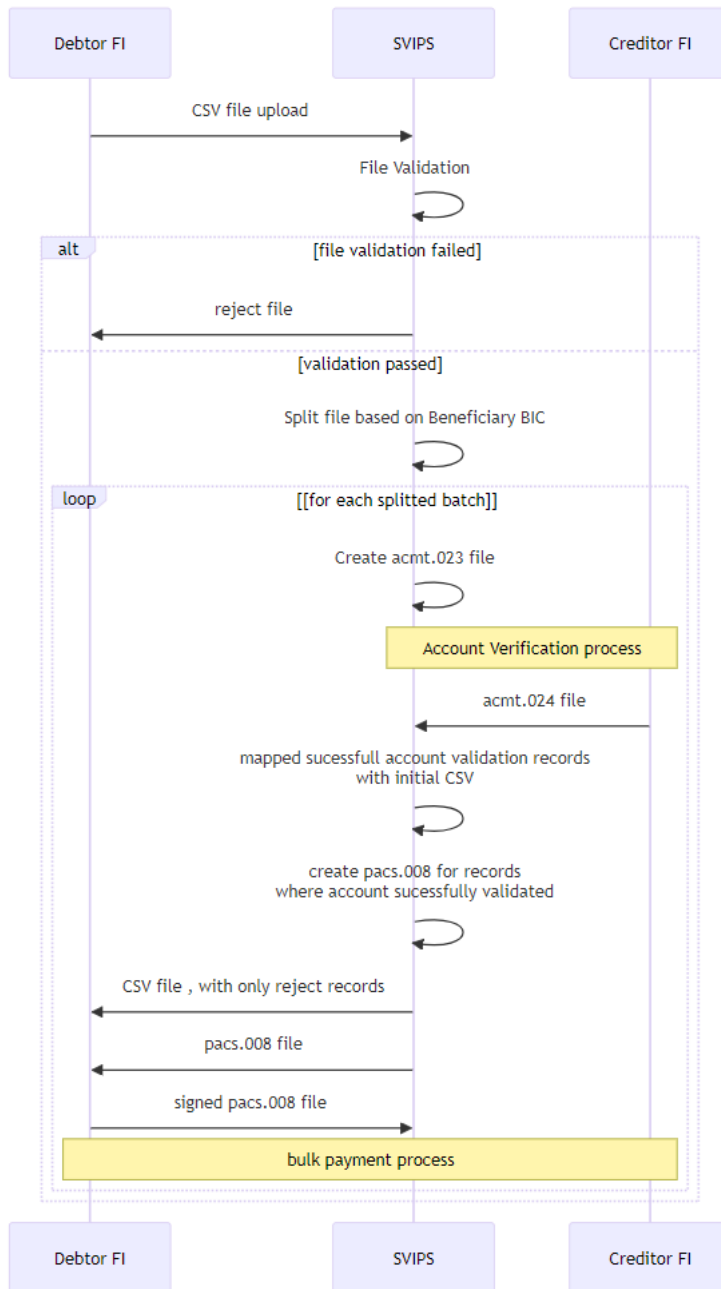


Figure 13 CSV file upload

2.5.1.1. CSV validation

File structure Check: all fields are present and in correct format.

Duplication check:

- Based on Batch ID
- Based on transaction ID

Note in the case of any error, the whole CSV file will be rejected.

2.6. Message and specification

2.6.1. Acmt.023

The IdentificationVerificationRequest message is sent before the sending of one or several transactions messages. The IdentificationVerificationRequest message can contain one or more verification requests.

The IdentificationVerificationRequestV04 MessageDefinition will be composed of 2 MessageBuildingBlocks:

A. Assignment

Identifies the identification assignment.

Or	MessageElement<XML Tag>	Mult.	Type	Constr. No.	Page
	MessageId Identification <MsgId>	[1..1]	Text		29
	CreationDate Time <CreDtTm>	[1..1]	DateTime		29
	Creator <Cretr>	[0..1]			29
{Or	Party <Pty>	[1..1]	±		30
	Assigner <Assgnr>	[1..1]			32
{Or	Party <Pty>	[1..1]	±		32
	Assignee <Assgne>	[1..1]			34
{Or	Party <Pty>	[1..1]	±		34

B. Verification

Presence: [1..*]

Information concerning the identification data that is requested to be verified.

Or	MessageElement<XML Tag>	Mult.	Type	Constr. No.	Page
	Identification <Id>	[1..1]	Text		36
	PartyAndAccount Identification <PtyAndAcctId>	[1..1]			36
	Party <Pty>	[0..1]	±		37
	Account <Acct>	[0..1]	±	C7, C6	38

2.6.2. Acmt.024

- The IdentificationVerificationReport message is sent as a response to an IdentificationVerificationRequest message.
- The IdentificationVerificationReport message can contain one or more reports.
- The IdentificationVerificationReport message may include a reason if the presented party and/or account identification information is confirmed to be incorrect.

The IdentificationVerificationReportV04 MessageDefinition will be composed of 3 MessageBuildingBlocks:

Assignment

Identifies the identification assignment.

Or	MessageElement<XML Tag>	Mult.	Type	Constr. No.	Page
	MessageIdentification <MsgId>	[1..1]	Text		29
	CreationDateTime <CreDtTm>	[1..1]	DateTime		29
	Creator <Cretr>	[0..1]			29
{Or	Party <Pty>	[1..1]	±		30
	Assigner <Assgnr>	[1..1]			32
{Or	Party <Pty>	[1..1]	±		32
	Assignee <Assgne>	[1..1]			34
{Or	Party <Pty>	[1..1]	±		34

OriginalAssignment

Provides for the reference to the original identification assignment.

OriginalAssignment <OrgnlAssgnmt> contains the following elements

Or	MessageElement<XML Tag>	Mult.	Type	Constr. No.	Page
	MessageIdentification <MsgId>	[0..1]	Text		52
	CreationDateTime <CreDtTm>	[0..1]	DateTime		52
	FirstAgent <FrstAgt>	[0..1]	±		53

Report

Presence: [1..*]

Information concerning the verification of the identification data for which verification was requested.

Or	MessageElement<XML Tag>	Mult.	Type	Constr. No.	Page
	OriginalIdentification <OrgnId>	[1..1]	Text		54
	Verification <Vrfctn>	[1..1]	Indicator		54
	Reason <Rsn>	[0..1]			54
{Or	Code <Cd>	[1..1]	CodeSet		55
Or}	Proprietary <Prtry>	[1..1]	Text		55
	OriginalPartyAndAccountIdentification <OrgnIPtyAndAcctId>	[0..1]			55
	Party <Pty>	[0..1]	±		55
	Account <Acct>	[0..1]	±	C7, C6	56
	Agent <Agt>	[0..1]	±		57

Proprietary code will be use to send the account verification status

STAC	Account Closed
STNT	Transaction not allowed
STAE	Account Does not Exist

2.6.3. Pacs.002

For batch processing the pacs.002 will contain only information at group level.

A. GroupHeader

Set of characteristics shared by all individual transactions included in the status report message.

Or	MessageElement<XML Tag>	Mult.	Type	Constr. No.	Page
	MessageIdentification <MsgId>	[1..1]	Text		12
	CreationDateTime <CreDtTm>	[1..1]	DateTime		12

B. OriginalGroupInformationAndStatus

Original group information concerning the group of transactions, to which the status report message refers to.

Or	MessageElement<XML Tag>	Mult.	Type	Constr. No.	Page
	OriginalMessageIdentification <OrgnlMsgId>	[1..1]	Text		14
	OriginalMessageNameIdentification <OrgnlMsgNmId>	[1..1]	Text		14
	OriginalCreationDateTime <OrgnlCreDtTm>	[0..1]	DateTime		14
	OriginalNumberOfTransactions <OrgnlNbOfTx>	[0..1]	Text		15
	OriginalControlSum <OrgnlCtrlSum>	[0..1]	Quantity		15
	GroupStatus <GrpSts>	[0..1]	CodeSet		15
	StatusReasonInformation <StsRsnInf>	[0..*]	±	C26	15
	NumberOfTransactionsPerStatus <NbOfTxPerSts>	[0..*]	±		15

3. Not Functional Requirement

3.1. Testing

Testing will be performed at the following levels:

1. System integration testing
2. User acceptance testing
3. Performance testing

4. References

4.1. BPC Documents

#	Reference	File Name	Provider

Table 2. BPC Documents

4.2. NSSL Documents

#	Reference	File Name	Provider

Table 3. NSSL Documents

4.3. External interface specifications

#	Reference	Document Name	Provider

Table 4. External interface specifications